ProcessTensors.jl: An ITensor-native framework for non-Markovian open quantum dynamics

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Abstract

We present ProcessTensors.jl, a Julia package for learning, constructing, and using process tensors with ITensor. The paper introduces process tensors from the ground up and connects the equations to tensor diagrams, tagged ITensor indices, and working code. Through guided examples, users can build process tensors, apply different sequences of quantum instruments, use testers with ancillary systems, and inspect memory across the temporal bonds. We implement exact and compressed ACE constructions and benchmark them against established software. We also review the current process-tensor software landscape and explain how the package can be extended with further PT-MPO construction methods, including ideas based on TEMPO, TEDOPA, and periodic process tensors. Our goal is to make process tensors approachable for new users while providing a clear foundation for future development of this package.

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Contents

1 Introduction

In principle, describing an open quantum system is easy: write down a giant wavefunction for the entire universe, place it in an even more gigantic Hilbert space, and evolve it according to the Schrödinger equation. Somewhere inside that wavefunction lies everything one could wish to know about the system of interest. In practice, of course, we cannot keep track of every surrounding particle, lattice vibration, stray electromagnetic field, or experimental imperfection and calculate their joint correlated dynamics. We therefore do what physicists have always done when the universe becomes inconveniently large: draw a boundary. The degrees of freedom we wish to study are called the *system*; everything beyond this boundary is called the *environment* or *bath*. Open quantum-system theory asks how the bath affects the system without requiring us to simulate the bath in all its microscopic detail.

The most familiar answers we would see in this regard are those that rely on weak-coupling, short-memory, or related approximations on the baths that lead to time-local master equations. In the Markovian limit, the system does not need a record of its earlier history in order to determine what happens next. Markovian master equations provide an extremely successful workhorse for open quantum dynamics, with the Gorini–Kossakowski–Sudarshan–Lindblad form supplying the standard generator of quantum dynamical semigroups [1,2]. Nevertheless, weak coupling and a clean separation of timescales are not available in every problem. Structured reservoirs, strong system–bath coupling, slow environmental modes, and delayed feedback can all preserve information about earlier dynamics [3,4]. In these regimes, memory is part of the physics rather than a correction that can safely be ignored.

The difficulty is not merely that the environment remembers, but that the system may give us no indication of *what* it remembers if we trace out the bath. To see this, consider two joint system–environment states at some time t_1,

$$\rho_{SE}^{(1)}(t_1) \qquad \text{and} \qquad \rho_{SE}^{(2)}(t_1), \tag{1}$$

which reduce to exactly the same system state,

$$\mathrm{Tr}_E\Big[\rho_{SE}^{(1)}(t_1)\Big] = \mathrm{Tr}_E\Big[\rho_{SE}^{(2)}(t_1)\Big] = \rho_S(t_1). \tag{2}$$

As far as the system density matrix is concerned, these two situations are indistinguishable. The environment, however, may disagree. It can retain different states or different correlations with the system in the two cases. Consequently, evolving both joint states with the same system–environment unitary need not produce the same reduced state at a later time:

$$\mathrm{Tr}_E\Big[U_{2:1}\rho_{SE}^{(1)}(t_1)U_{2:1}^{\dagger}\Big] \neq \mathrm{Tr}_E\Big[U_{2:1}\rho_{SE}^{(2)}(t_1)U_{2:1}^{\dagger}\Big]. \tag{3}$$

Thus, even perfect knowledge of the present system state need not be sufficient to predict its future. The information distinguishing the two experiments has simply moved somewhere beyond the boundary we drew earlier.

The limitation becomes sharper when we intervene our system along the way. Imagine performing two experiments with different histories and, at an intermediate time t_1, discarding the system state and preparing the same fresh state P in both. This operation—often called a *causal break*—removes any information that could pass directly from the past to the future through the system. Immediately afterward, both experiments therefore contain the same reduced state,

$$\rho_S^{(1)}(t_1^+) = \rho_S^{(2)}(t_1^+) = P. \tag{4}$$

Nevertheless, the intervention acts on a system that may already be correlated with its environment. A measurement outcome, for example, can condition the environment into a corresponding state, while repreparing P makes the system look identical in every run. If the later dynamics still depends on the earlier preparation or measurement outcome, then the memory must have travelled across the causal break through the environment. Such memory can remain hidden in an uninterrupted reduced trajectory, and can even occur when familiar two-time criteria describe the unperturbed evolution as Markovian [5].

This leaves a state-to-state description with an awkward question. What map should take P at t_1 to the state at t_2, if the same P can lead to different futures depending on what happened before it was prepared? There is no contradiction in quantum mechanics here; rather, we have asked the reduced state to remember information that it no longer contains. The missing input is the history of interventions—the preparations, controls, and measurements through which the system and its environment arrived at the present experiment. A complete description of the quantum process must therefore accept an entire ordered sequence of operations, rather than only a density matrix at one particular time.

The process tensor provides precisely such a description [6,7]. For a sequence of interventions $(\mathcal{A}_0, \mathcal{A}_1, \ldots, \mathcal{A}_{n-1})$, it returns the final system state,

$$\rho_S(t_n) = \mathcal{T}_{n:0}\big[\mathcal{A}_{n-1}, \ldots, \mathcal{A}_0\big], \tag{5}$$

or, when the interventions contain measurements, the corresponding outcome probabilities and conditional states. A reduced trajectory tells us what happened under one prescribed history; a process tensor tells us what could have happened had we intervened differently along the way. It does not reveal every microscopic bath degree of freedom, but retains their complete influence on experiments performed through the system (the mathematical construction behind this statement—including its Choi representation and temporal input and output spaces—will be developed carefully in Sec. 4. For the present discussion, its operational meaning is enough). This makes causal breaks, multi-time correlations, instrument sequences, Markov order, and correlations across temporal partitions accessible within one description. In the Markovian limit, the process separates into independent transformations between successive times; otherwise, its temporal parts remain correlated, recording how information stored in the environment can return later. Memory thus becomes part of the object being manipulated rather than a feature inferred from a single trajectory.

The process tensor provides precisely such a description [6,7]. For a sequence of interventions $(\mathcal{A}_0, \mathcal{A}_1, \ldots, \mathcal{A}_{n-1})$, it returns the final system state,

$$\rho_S(t_n) = \mathcal{T}_{n:0}\big[\mathcal{A}_{n-1}, \ldots, \mathcal{A}_0\big], \tag{6}$$

or, when the interventions contain measurements, the corresponding outcome probabilities and conditional states. A reduced trajectory tells us what happened when one preparation was allowed to evolve under one prescribed history. A process tensor tells us what could have happened had we prepared, controlled, or measured the system differently along the way. The mathematical construction behind this statement—including its Choi representation and temporal input and output spaces—will be developed carefully in Sec. 4. For the present discussion, its operational meaning is enough: the process tensor describes a multi-time experiment rather than a collection of isolated snapshots. This does not mean that the process tensor opens the bath and presents every microscopic oscillator for inspection. Different microscopic environments can produce the same observable influence on the system and therefore the same process tensor. Instead, it retains everything about the unresolved environment that can affect future experiments performed through the system. Environmental memory can then be studied through causal breaks, multi-time correlations, different instrument sequences, operational notions of Markov order, and correlations across temporal partitions. In the Markovian limit, the process separates into independent transformations between successive times. Away from that limit, the temporal pieces remain correlated, recording how information placed in the environment at an earlier time can return during a later intervention. Memory is no longer inferred only from an unusual feature in one trajectory; it becomes part of the mathematical object being manipulated.

2 Liouville-space foundations

2.1 Vectorisation of operators

Use sections to structure your article's presentation.

Equations should be centered; multi-line equations should be aligned.

$$H=\sum_{j=1}^{N}\Big[J(S_j^x S_{j+1}^x+S_j^y S_{j+1}^y+\Delta S_j^z S_{j+1}^z)-hS_j^z\Big]. \tag{7}$$

All equations and references should be hyperlinked to ensure ease of navigation. This also holds for [sub]sections: readers should be able to easily jump to Section 3.

2.2 Superoperators, Liouvillians, and channels

2.3 From spatial to temporal tensor networks

3 Software architecture

There is no strict length limitation, but the authors are strongly encouraged to keep contents to the strict minimum necessary for peers to reproduce the research described in the paper.

3.1 ITensor as the tensor-network substrate

You are free to use dividers as you see fit.

3.2 Hilbert- and Liouville-space objects

Figures should only occupy the stricly necessary space, in any case individually fitting on a single page. Each figure item should be appropriately labeled and accompanied by a descriptive caption. **SciPost does not accept creative or promotional figures or artist's impressions**; on the other hand, technical drawings and scientifically accurate representations are encouraged.

3.3 Systems and environments

3.4 A backend-oriented process-tensor API

4 Process tensors as first-class tensor-network objects

4.1 From a channel to a process

4.2 Anatomy of `ProcessTensor`

4.3 Building a small process tensor

4.4 Construction backends: dense reference and ACE

5 Quantum instruments

5.1 Operational meaning

5.2 Instrument organisation in `ProcessTensors.jl`

5.3 `InstrumentSeq`: a temporal analogue of `OpSum`

6 Contracting and evaluating processes

6.1 General contraction with `evaluate_process`

6.2 Reduced dynamics with `evolve`

6.3 Open process legs

7 Common process-tensor workflows

7.1 Final reduced density matrices

7.2 Observable expectation values

7.3 Probabilities and trace-non-preserving branches

7.4 Multi-time correlation functions

8 Testers and enlarged interventions

Status: planned for the paper release; API under development.

9 Characterising process tensors and quantum memory

Section lead: Aaron Sander. The text below is a structural placeholder and should not be read as claiming currently released package functionality.

10 Validation and ACE benchmarks

10.1 Correctness against dense and exact references

10.2 Compression accuracy and temporal bond dimension

10.3 Runtime and memory scaling

10.4 Comparison with the C++ ACE toolkit

11 Conclusion and outlook

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Acknowledgements

Acknowledgements should follow immediately after the conclusion.

Author contributions This is optional. If desired, contributions should be succinctly described in a single short paragraph, using author initials.

Funding information Authors are required to provide funding information, including relevant agencies and grant numbers with linked author's initials. Correctly-provided data will be linked to funders listed in the Fundref registry.

Code and data availability The development repository is available at https://github.com/Gauthameshwar/ProcessTensors.jl and the documentation at https://gauthameshwar.github.io/ProcessTensors.jl/stable/. The final paper will cite an immutable release and archive the scripts/data used for all benchmark figures. **[TODO: insert release DOI/Zenodo or SciPost codebase DOI at submission.]**

A Broader Hilbert- and Liouville-space functionality

Add material which is better left outside the main text in a series of Appendices labeled by capital letters.

B Vectorisation conventions and index orientation

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